

Appendix

Project Title: Student Dropout Prediction

Course: BANA 288 PREDICTIVE ANALYTICS

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Group 1

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1. Overview

This appendix provides detailed technical documentation of the analytical process used in this project on predicting student dropout. It includes data preparation and preprocessing steps, descriptive analysis of key variables, development of predictive models (logistic regression and decision trees), and model evaluation using multiple performance metrics. In addition, it presents the application of the final model to currently enrolled students to identify those at high risk of dropout. This document is intended to complement the presentation by providing transparency, reproducibility, and additional analytical details.

2. Data Preparation & Manipulation

The dataset was obtained from the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>) and contains student records from the Polytechnic Institute of Portalegre between 2008 and 2019.

The original dataset includes 4,424 undergraduate students and 36 variables covering demographic, socioeconomic, academic, and macroeconomic factors.

To prepare the data for analysis, students with the status "Enrolled" were removed, as their final outcomes are unknown. This resulted in a dataset with 3,630 observations used for modeling.

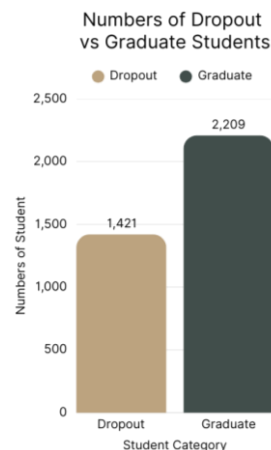
In addition, variables that directly reflect academic performance, such as grades, curricular units, and special needs indicators, were excluded to focus on early and indirect predictors of dropout. This allows the model to identify at-risk students before academic performance declines. Categorical variables were recoded into more interpretable labels (e.g., converting

binary values such as 0/1 into “Female/Male” or “Daytime/Evening”) to improve interpretability.

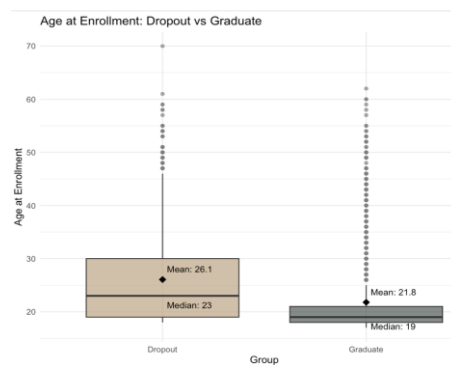
3. Descriptive Analysis

Descriptive analysis was conducted to understand the distribution of student outcomes and key variables.

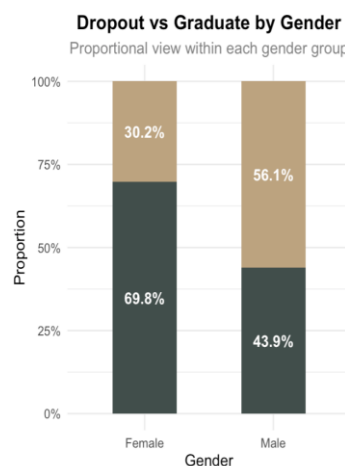
The overall distribution shows that the number of graduate students exceeds the number of dropout students, indicating a moderately imbalanced dataset.



Age at enrollment was analyzed using summary statistics and boxplots. The results suggest noticeable differences in central tendency between the two groups, with dropout and graduate students exhibiting different age patterns.



Additional analysis examined the distribution of gender and attendance type. While gender distribution is relatively balanced, further breakdown by outcome reveals differences in dropout proportions within each group.



4. Predictive Modeling Approach

To investigate the factors associated with student dropout, two predictive modeling approaches were implemented: logistic regression and decision tree models.

To better understand the contribution of different types of variables, the dataset was divided into four groups: demographic, academic, socioeconomic, and macroeconomic variables. Separate models were built for each group, as well as a combined model including all variables.

Table 1. Variable Groups Used for Model Development

Variable Group	Variables Included
Demographic	Marital Status, Nationality, Displaced, Gender, Age at Enrollment, International
Academic	Application Mode, Application Order, Attendance Time
Socioeconomic	Mother's Qualification, Father's Qualification, Mother's Occupation, Father's Occupation, Debtor
Macroeconomic	Unemployment Rate, Inflation Rate, GDP
Combined Model	All variables listed above

To better understand these variables, below is the variables table:

Table 2. Variables Table

Variable Name	Role	Type	Description	Missing Value
Marital Status	Feature	Integer	1 – single 2 – married 3 – widower 4 – divorced 5 – facto union 6 – legally separated	No
Nationality	Feature	Integer	1 - Portuguese; 2 - German; 6 - Spanish; 11 - Italian; 13 - Dutch; 14 - English; 17 - Lithuanian; 21 - Angolan; 22 - Cape Verdean; 24 - Guinean; 25 - Mozambican; 26 - Santomean; 32 - Turkish; 41 - Brazilian; 62 - Romanian; 100 - Moldova (Republic of); 101 - Mexican; 103 - Ukrainian; 105 - Russian; 108 - Cuban; 109 - Colombian	No
Displaced	Feature	Integer	1 – yes 0 – no	No
Gender	Feature	Integer	1 – male 0 – female	No
Age at enrollment	Feature	Integer	Age of student at enrollment	No
International	Feature	Integer	1 – yes 0 – no	No

Variable Name	Role	Type	Description	Missing Value
Application mode	Feature	Integer	1 - 1st phase - general contingent 2 - Ordinance No. 612/93 5 - 1st phase - special contingent (Azores Island) 7 - Holders of other higher courses 10 - Ordinance No. 854-B/99 15 - International student (bachelor) 16 - 1st phase - special contingent (Madeira Island) 17 - 2nd phase - general contingent 18 - 3rd phase - general contingent 26 - Ordinance No. 533-A/99, item b2) (Different Plan) 27 - Ordinance No. 533-A/99, item b3 (Other Institution) 39 - Over 23 years old 42 - Transfer 43 - Change of course 44 - Technological specialization diploma holders 51 - Change of institution/course 53 - Short cycle diploma holders 57 - Change of institution/course (International)	No
Application order	Feature	Integer	Application order (between 0 - first choice; and 9 last choice)	No
Attendance Time	Feature	Integer	1 – daytime 0 - evening	No
Mother's qualification	Feature	Integer	1 - Secondary Education - 12th Year of Schooling or Eq. 2 - Higher Education - Bachelor's Degree 3 - Higher Education - Degree 4 - Higher Education - Master's 5 - Higher Education - Doctorate 6 - Frequency of Higher Education 9 - 12th Year of Schooling - Not Completed 10 - 11th Year of Schooling - Not Completed 11 - 7th Year (Old) 12 - Other - 11th Year of Schooling 14 - 10th Year of Schooling 18 - General commerce course 19 - Basic Education 3rd Cycle (9th/10th/11th Year) or Equiv. 22 - Technical-professional course 26 - 7th year of schooling 27 - 2nd cycle of the general high school course 29 - 9th Year of Schooling - Not Completed 30 - 8th year of schooling 34 - Unknown	No

Variable Name	Role	Type	Description	Missing Value
			35 - Can't read or write 36 - Can read without having a 4th year of schooling 37 - Basic education 1st cycle (4th/5th year) or equiv. 38 - Basic Education 2nd Cycle (6th/7th/8th Year) or Equiv. 39 - Technological specialization course 40 - Higher education - degree (1st cycle) 41 - Specialized higher studies course 42 - Professional higher technical course 43 - Higher Education - Master (2nd cycle) 44 - Higher Education - Doctorate (3rd cycle)	
Father's qualification	Feature	Integer	1 - Secondary Education - 12th Year of Schooling or Eq. 2 - Higher Education - Bachelor's Degree 3 - Higher Education - Degree 4 - Higher Education - Master's 5 - Higher Education - Doctorate 6 - Frequency of Higher Education 9 - 12th Year of Schooling - Not Completed 10 - 11th Year of Schooling - Not Completed 11 - 7th Year (Old) 12 - Other - 11th Year of Schooling 13 - 2nd year complementary high school course 14 - 10th Year of Schooling 18 - General commerce course 19 - Basic Education 3rd Cycle (9th/10th/11th Year) or Equiv. 20 - Complementary High School Course 22 - Technical-professional course 25 - Complementary High School Course - not concluded 26 - 7th year of schooling 27 - 2nd cycle of the general high school course 29 - 9th Year of Schooling - Not Completed 30 - 8th year of schooling 31 - General Course of Administration and Commerce 33 - Supplementary Accounting and Administration 34 - Unknown 35 - Can't read or write 36 - Can read without having a 4th year of schooling 37 - Basic education 1st cycle (4th/5th year) or equiv. 38 - Basic	No

Variable Name	Role	Type	Description	Missing Value
			Education 2nd Cycle (6th/7th/8th Year) or Equiv. 39 - Technological specialization course 40 - Higher education - degree (1st cycle) 41 - Specialized higher studies course 42 - Professional higher technical course 43 - Higher Education - Master (2nd cycle) 44 - Higher Education - Doctorate (3rd cycle)	
Mother's occupation	Feature	Integer	0 - Student 1 - Representatives of the Legislative Power and Executive Bodies, Directors, Directors and Executive Managers 2 - Specialists in Intellectual and Scientific Activities 3 - Intermediate Level Technicians and Professions 4 - Administrative staff 5 - Personal Services, Security and Safety Workers and Sellers 6 - Farmers and Skilled Workers in Agriculture, Fisheries and Forestry 7 - Skilled Workers in Industry, Construction and Craftsmen 8 - Installation and Machine Operators and Assembly Workers 9 - Unskilled Workers 10 - Armed Forces Professions 90 - Other Situation 99 - (blank) 122 - Health professionals 123 - teachers 125 - Specialists in information and communication technologies (ICT) 131 - Intermediate level science and engineering technicians and professions 132 - Technicians and professionals, of intermediate level of health 134 - Intermediate level technicians from legal, social, sports, cultural and similar services 141 - Office workers, secretaries in general and data processing operators 143 - Data, accounting, statistical, financial services and registry-related operators 144 - Other administrative support staff 151 - personal service workers 152 - sellers 153 - Personal care workers	No

Variable Name	Role	Type	Description	Missing Value
			and the like 171 - Skilled construction workers and the like, except electricians 173 - Skilled workers in printing, precision instrument manufacturing, jewelers, artisans and the like 175 - Workers in food processing, woodworking, clothing and other industries and crafts 191 - cleaning workers 192 - Unskilled workers in agriculture, animal production, fisheries and forestry 193 - Unskilled workers in extractive industry, construction, manufacturing and transport 194 - Meal preparation assistants	
Father's occupation	Feature	Integer	0 - Student 1 - Representatives of the Legislative Power and Executive Bodies, Directors, Directors and Executive Managers 2 - Specialists in Intellectual and Scientific Activities 3 - Intermediate Level Technicians and Professions 4 - Administrative staff 5 - Personal Services, Security and Safety Workers and Sellers 6 - Farmers and Skilled Workers in Agriculture, Fisheries and Forestry 7 - Skilled Workers in Industry, Construction and Craftsmen 8 - Installation and Machine Operators and Assembly Workers 9 - Unskilled Workers 10 - Armed Forces Professions 90 - Other Situation 99 - (blank) 101 - Armed Forces Officers 102 - Armed Forces Sergeants 103 - Other Armed Forces personnel 112 - Directors of administrative and commercial services 114 - Hotel, catering, trade and other services directors 121 - Specialists in the physical sciences, mathematics, engineering and related techniques 122 - Health professionals 123 - teachers 124 - Specialists in finance, accounting, administrative organization, public and commercial	No

Variable Name	Role	Type	Description	Missing Value
			relations 131 - Intermediate level science and engineering technicians and professions 132 - Technicians and professionals, of intermediate level of health 134 - Intermediate level technicians from legal, social, sports, cultural and similar services 135 - Information and communication technology technicians 141 - Office workers, secretaries in general and data processing operators 143 - Data, accounting, statistical, financial services and registry-related operators 144 - Other administrative support staff 151 - personal service workers 152 - sellers 153 - Personal care workers and the like 154 - Protection and security services personnel 161 - Market-oriented farmers and skilled agricultural and animal production workers 163 - Farmers, livestock keepers, fishermen, hunters and gatherers, subsistence 171 - Skilled construction workers and the like, except electricians 172 - Skilled workers in metallurgy, metalworking and similar 174 - Skilled workers in electricity and electronics 175 - Workers in food processing, woodworking, clothing and other industries and crafts 181 - Fixed plant and machine operators 182 - assembly workers 183 - Vehicle drivers and mobile equipment operators 192 - Unskilled workers in agriculture, animal production, fisheries and forestry 193 - Unskilled workers in extractive industry, construction, manufacturing and transport 194 - Meal preparation assistants 195 - Street vendors (except food) and street service providers	
Debtor	Feature	Integer	1 – yes 0 – no	No
Unemployment	Feature	Continuous	Unemployment rate (%)	No

Variable Name	Role	Type	Description	Missing Value
rate				
Inflation rate	Feature	Continuous	Inflation rate (%)	No
GDP	Feature	Continuous	GDP	No
Target	Target	Categorical	Target. The problem is originally formulated as a three category classification task (dropout, enrolled, and graduate) at the end of the normal duration of the course. *During model training, the "Enrolled" category is temporarily excluded, and the analysis is conducted as a binary classification problem (dropout vs. graduate).	No

Logistic regression models were developed to estimate the probability of dropout. Model performance was evaluated using deviance reduction as a measure of explanatory power. The combined model using all indirect variables achieved the highest deviance reduction (14.8%), outperforming models built on individual variable groups. Among the subsets, demographic variables showed relatively strong predictive power, followed by socioeconomic and academic variables, while macroeconomic variables contributed minimally.

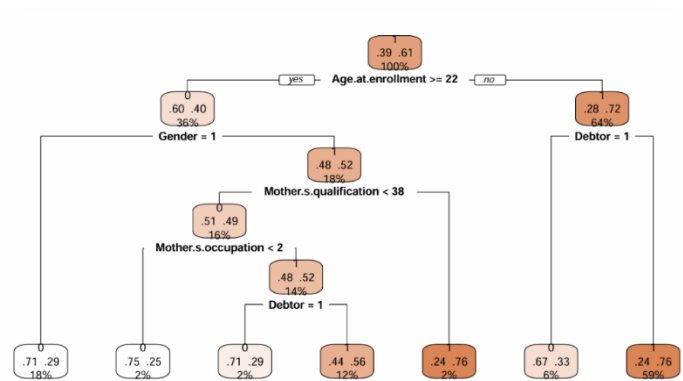
Table 3. Significant Predictors from Logistic Regression

Variable	Coefficient	P-value	Category
Application Mode	-0.015	<0.001	Academic
Daytime/Evening Attendance	-0.388	0.048	Academic
Debtor	-1.677	<0.001	Socioeconomic
Gender	-0.893	<0.001	Demographic
Age at Enrollment	-0.063	<0.001	Demographic
Unemployment Rate	0.049	0.028	Macroeconomic
GDP	0.054	0.039	Macroeconomic

In addition to logistic regression, decision tree models were constructed to capture non-linear relationships and interactions among variables. Separate trees were built for each variable group, as well as a combined tree model.

The combined decision tree identified age at enrollment as the primary splitting variable, followed by financial and demographic factors such as debtor status and gender. This suggests that student dropout is influenced by a combination of demographic and socioeconomic

factors rather than a single variable.



5. Model Evaluation and Comparison

To evaluate model performance, both logistic regression and decision tree models were tested on the hold-out test dataset. Several evaluation metrics were used, including accuracy, precision, recall, F1-score, and area under the ROC curve (AUC).

Table 4. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	0.696	0.705	0.860	0.775	0.752
Decision Tree	0.704	0.751	0.769	0.760	0.690

The logistic regression model achieved higher overall performance across most metrics compared to the decision tree model. In particular, it demonstrated a higher recall and AUC value, indicating a stronger ability to correctly identify dropout students and better distinguish between dropout and graduate outcomes.

Confusion matrices were constructed for both models to further examine classification performance. The logistic regression model showed a more balanced trade-off between precision and recall, while the decision tree model tended to be less stable and more sensitive to data variation.

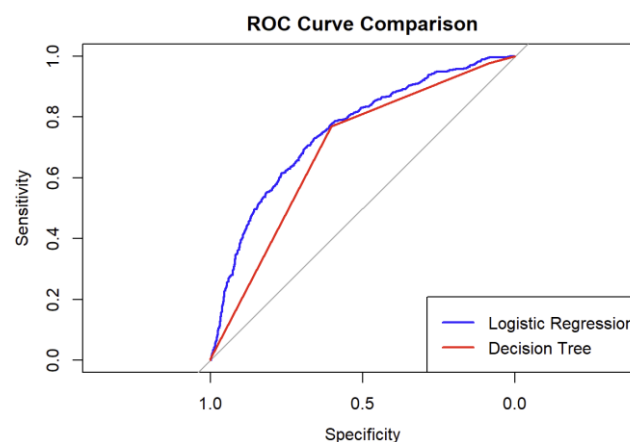
Table 5. Confusion Matrix – Logistic Regression

	Actual: Dropout (0)	Actual: Graduate (1)
Predicted 0	314	155
Predicted 1	397	950

Table 6. Confusion Matrix – Decision Tree

	Actual: Dropout (0)	Actual: Graduate (1)
Predicted 0	429	255
Predicted 1	282	850

ROC curves were also plotted to compare the models visually. The curve for logistic regression consistently lay above that of the decision tree, confirming its superior classification performance.



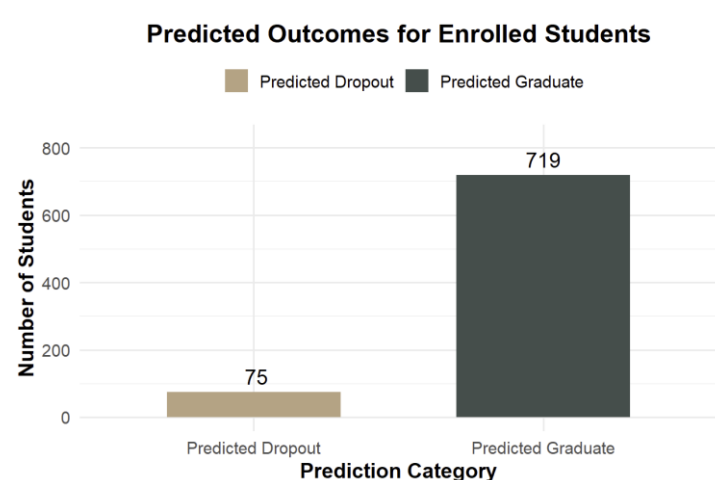
Based on these results, logistic regression was selected as the final model for predicting student outcomes, due to its higher predictive performance and more consistent behavior across evaluation metrics.

6. Real-World Application

To extend the analysis to a real-world application, the final logistic regression model was applied to students who are currently enrolled.

Using the trained model, the probability of graduation was predicted for each enrolled student. These probabilities were then converted into dropout risk scores. A threshold of 0.7 was used to classify students as high-risk of dropout.

Based on this threshold, students were categorized into two groups: predicted dropout and predicted graduate. The results provide a clear segmentation of the enrolled population, allowing identification of students who may require additional support.



This application demonstrates how predictive modeling can be used not only for analysis, but also for actionable decision-making. Building on these findings, several practical

recommendations can be proposed for universities. First, expanding financial support programs, such as scholarships and flexible monthly payment plans, may help reduce financial pressure for at-risk students. Second, early academic advising and tutoring services can support students who may struggle with coursework during the initial stages of their studies. Third, mentorship and engagement programs, including alumni outreach and peer support initiatives, can foster a stronger sense of belonging among students from diverse backgrounds.

In addition, predictive analytics should be integrated into institutional decision-making processes to proactively identify high-risk students. Continuous monitoring of indirect variables, such as socioeconomic and demographic factors, can further improve early warning systems and enable more targeted interventions.

7. Summary of Findings

This study aimed to identify key factors associated with student dropout and to develop predictive models to support early intervention strategies.

The analysis revealed that dropout is influenced by a combination of demographic, academic, macroeconomic and socioeconomic factors. In particular, variables such as age at enrollment, gender, and debtor status were found to be significant predictors, indicating that both individual characteristics and financial conditions play an important role in student outcomes. Academic-related variables also contributed to prediction, while macroeconomic factors showed relatively weaker effects.

Among the models evaluated, logistic regression demonstrated more consistent and reliable performance compared to the decision tree model, particularly in terms of recall and AUC. This suggests that logistic regression is more effective in identifying students at risk of dropout and provides better overall classification capability.

The application of the model to currently enrolled students further highlights its practical value. By identifying high-risk students early, institutions can implement targeted interventions to improve retention and support student success.

Overall, this project demonstrates how data-driven approaches can enhance decision-making in higher education, enabling universities to move from reactive strategies to proactive and preventive student support systems.

8. Limitations

While the analysis provides valuable insights into student dropout, several limitations should be acknowledged.

First, the dataset may not be fully representative of student populations in other countries or educational systems, which may limit the generalizability of the findings.

Second, the model does not account for certain external factors that can significantly influence dropout behavior, such as mental health conditions, personal motivation, or unexpected life events. These factors are difficult to quantify but may play an important role in student outcomes.

Third, early academic performance indicators were not included in the model. Incorporating such variables could potentially improve predictive accuracy, as academic struggles often emerge before dropout occurs.

Finally, the analysis primarily relies on indirect variables, which may not capture student behavior as effectively as more direct measures of academic engagement or performance.

These limitations suggest opportunities for future research and model improvement, including incorporating additional data sources and exploring more comprehensive predictors.